

# Task Analysis

## KLM

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# Mental models

- Characteristics of mental models
  - Analogies with familiar objects or domains
  - Fragmentary beliefs about parts of the system
  - May share the structure of the physical world they represent
  - Can be derived from perception or language/imagination
- Payoff
  - Improved problem solving—Mental models can furnish a problem space
- Limitations
  - Form of the information shapes the model, which shapes the kinds of tasks users can do

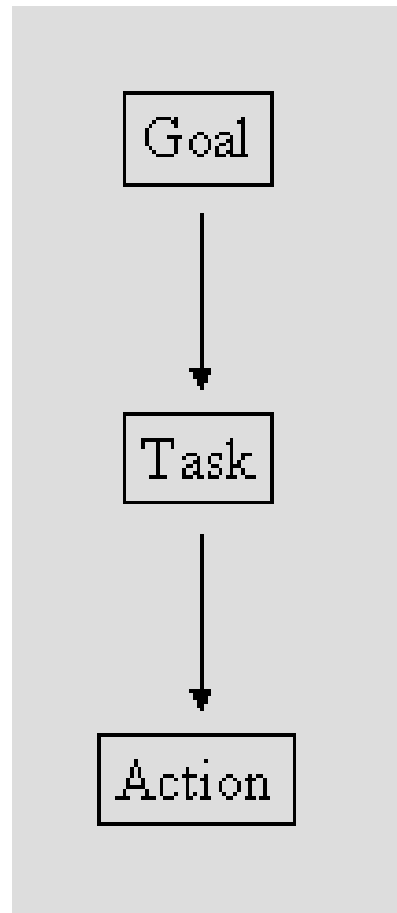
# Representations

- Representations are informationally equivalent with respect to a set of tasks if they allow the same tasks to be performed
- Representations are computationally equivalent with respect to the tasks if the relative difficulty of the tasks is the same.

# Task Analysis - What's a Task?

- A set of human actions that contributes to a functional objective and to the goal of the system.
- Scope or size of a task is determined by the definition of the objectives.
- Each task should be approximately equal in size.
  - But not always the case

# Task - Decomposition



# The difference is ...

- **Goal** - state of the system that a human wants to accomplish.
- **Task** - activities required, used, or deemed necessary to achieve a goal.
- **Actions** - steps required to complete the task.

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## What is Task Analysis?

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Methods of analysing people's jobs:

- what people do
- what things they work with
- what they must know

### **Example:**

in order to clean the house

- get the vacuum cleaner out
- fix the appropriate attachment
- clean the rooms
- when the dust bag gets full, empty it
- put the vacuum cleaner and tools away

Must know about:

vaccum cleaners, their attachments,  
dust bags, cupboards, rooms etc.

# Task Analysis vs.....

- Engineering requirements analysis defines performance required of hardware.
- Programming specs define performance of software.
- Task analysis defines performance of humans.



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## Approaches to task analysis

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- **Task decomposition**  
splitting task into (ordered) subtasks
- **Knowledge based techniques**  
what the user knows about the task  
and how it is organised
- **Entity–relation based analysis**  
relationships between objects and actions  
and the people who perform them

General method:

- *observe*  
unstructured lists of words and actions
- *organize*  
using notation or diagrams

# Task Decomposition

- Aims:
  - describe the actions people do
  - structure them within task subtask hierarchy
  - describe order of subtasks
- Focus on Hierarchical Task Analysis (HTA)
  - It uses:
    - text and diagrams to show hierarchy
    - plans to describe order

# Practical Task Analysis

- Develop concrete, detailed examples of tasks, users perform or want to perform.
- Determine what the user wants to do, not how to do it.
  - No assumptions about interface ??????
  - Allows design alternatives
  - Task descriptions are very specific
  - Task descriptions are context-specific
  - Task descriptions are user-specific

# Textual HTA description

- Hierarchy description ...

0. in order to clean the house

1. get the vacuum cleaner out

2. fix the appropriate attachment

3. clean the rooms

- 3.1. clean the hall

- 3.2. clean the living rooms

- 3.3. clean the bedrooms

4. empty the dust bag

5. put vacuum cleaner and attachments away

# Plans

- ... and plans
  - Plan 0: do 1 - 2 - 3 - 5 in that order. when the dust bag gets full do 4.
  - Plan 3: do any of 3.1, 3.2 or 3.3 in any order depending on which rooms need cleaning
- Note: only the plans denote order

# Generating the hierarchy

- get flat list of tasks
- group tasks into higher level tasks
- decompose lowest level tasks further

**Stopping rules** How do we know when to stop?

Is “*empty the dust bag*” simple enough?

**Purpose:** expand only relevant tasks.

**Error cost:** stop when  $P \times C$  is small

- Probability of making an error  $\times$  cost of the error

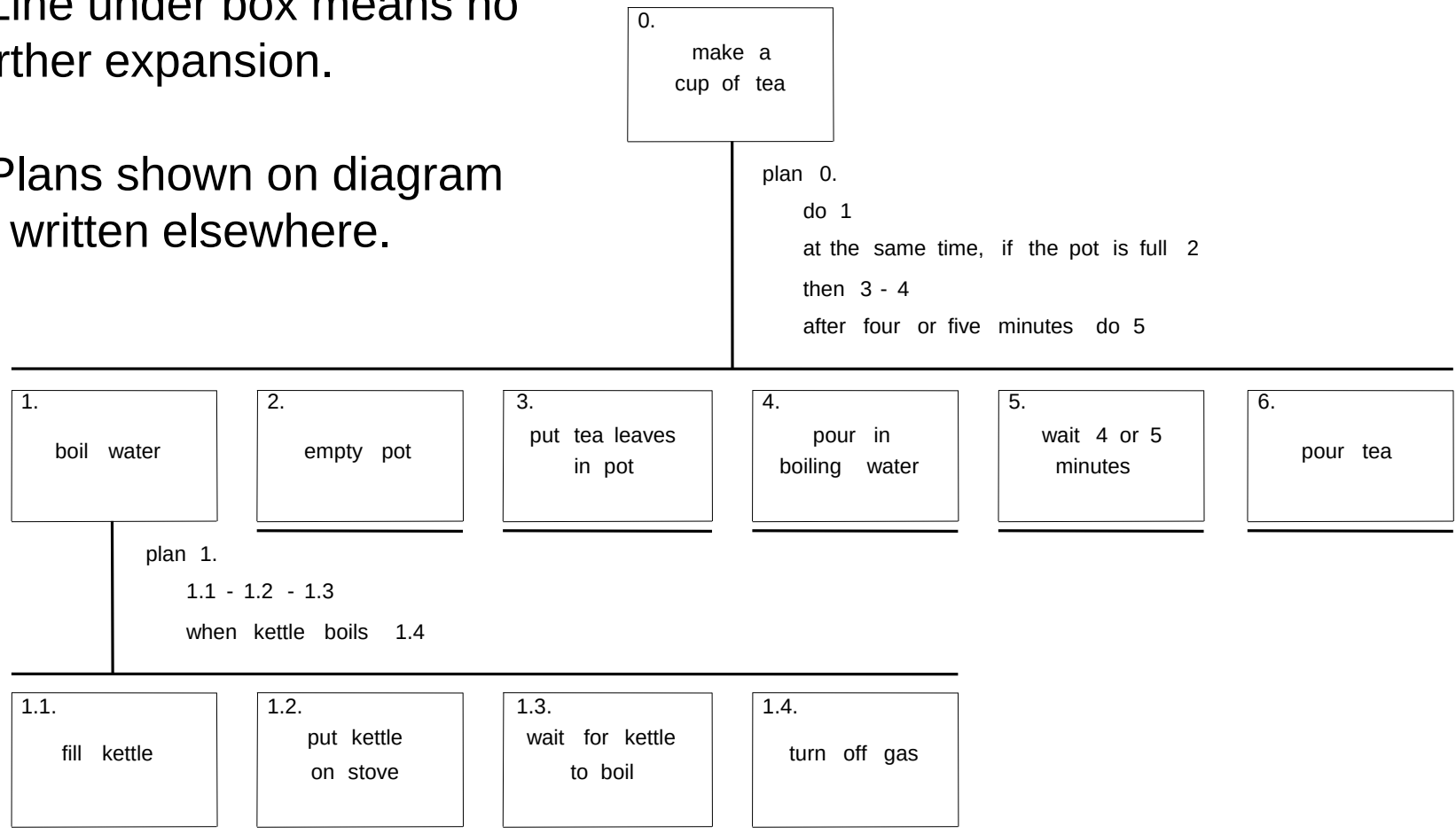
**Motor actions:** lowest sensible level

# Diagrammatic HT A

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- Line under box means no further expansion.

- Plans shown on diagram or written elsewhere.



# Refining the description

- Given initial HTA (textual or diagram)

How to check/improve it?

- Some heuristics:

- **paired actions**

e.g., where is 'turn on gas'

- **restructure**

e.g., generate task 'make pot'

- **balance**

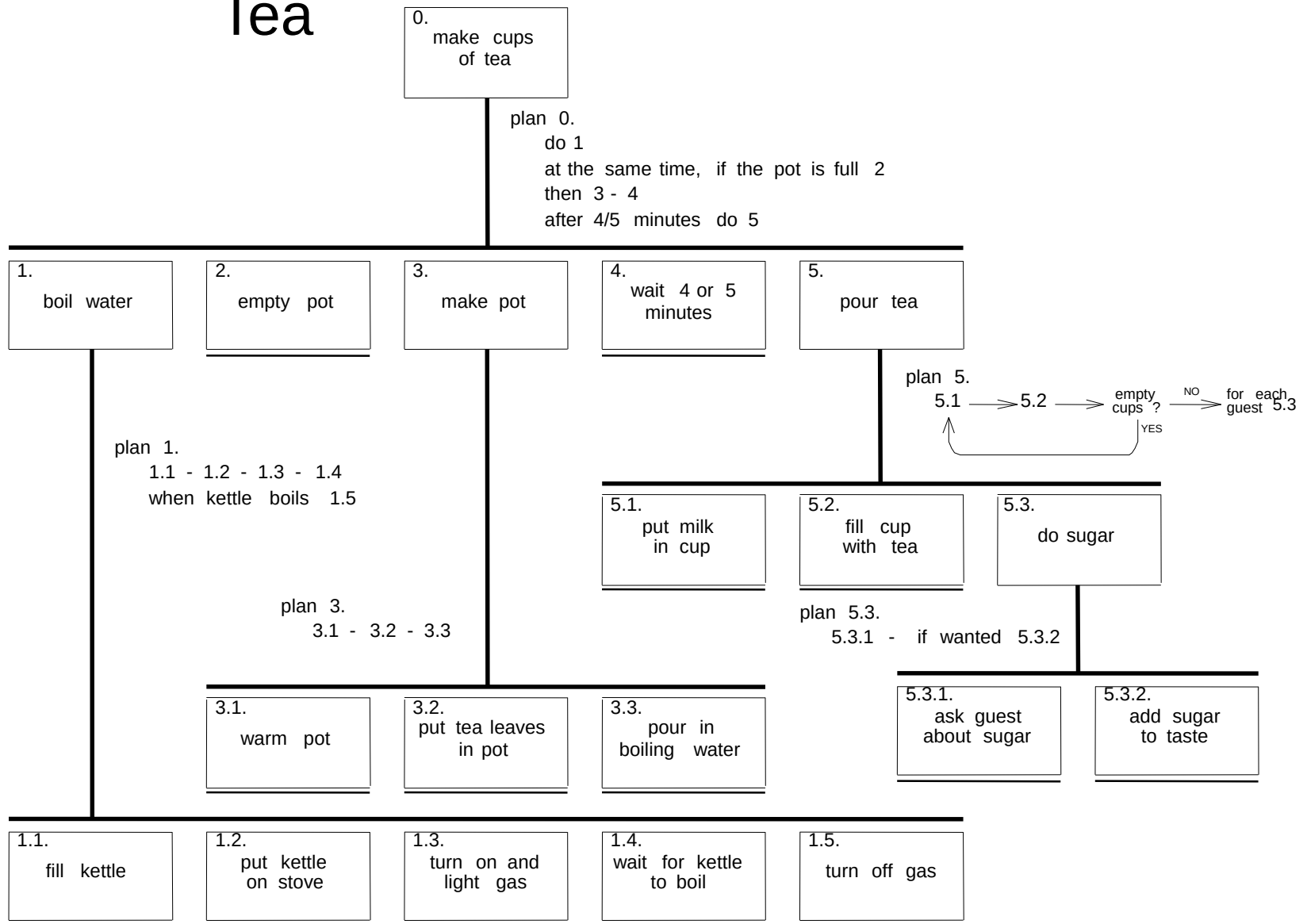
e.g., is 'pour tea' simpler than making pot?

- **generalize**

e.g., make one cup or two ... or more



# Redefined HTA For Making Tea



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## Types of plan

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### **fixed sequence**

e.g., 1.1–1.2–1.3

### **optional tasks**

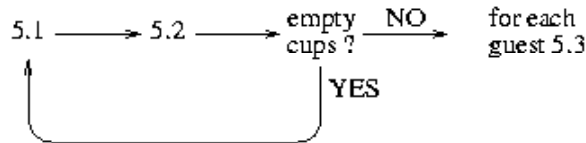
e.g., *if* the pot is full 2

### **waiting for events**

e.g., *when* kettle boils 1.4

### **cycles**

Plan 5.



### **time-sharing**

e.g., do 1; at the *same time* ...

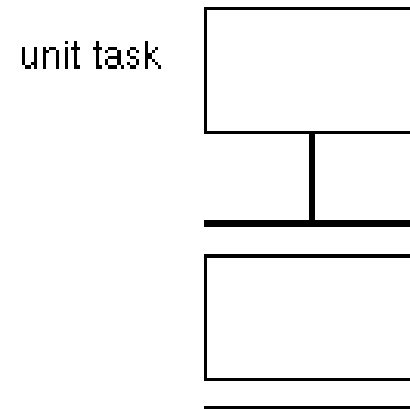
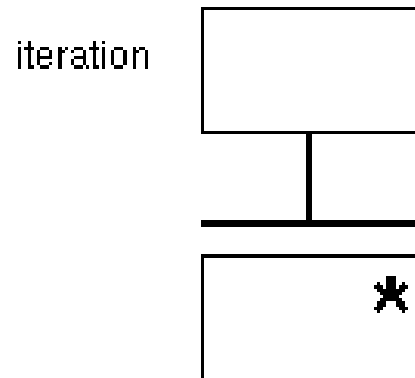
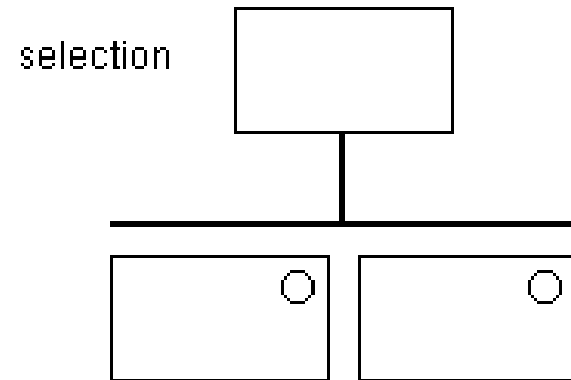
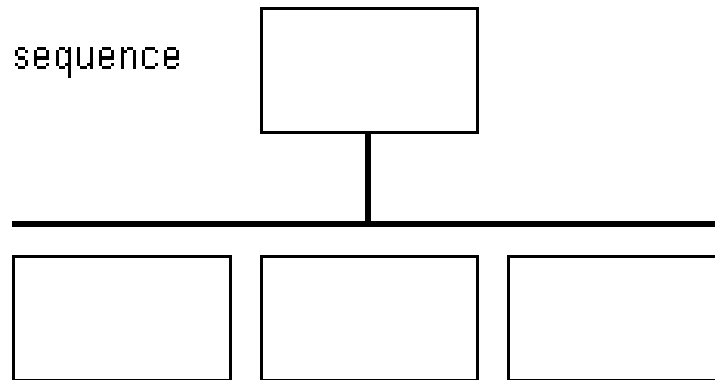
### **discretionary**

e.g., do any of 3.1, 3.2 or 3.3 in *any order*

### **mixtures**

most plans involve several of the above

# HTA Structure Chart Notation



# Stages of a HTA

## 1. Starting the analysis

- a) Specify the main task.
- b) Break down main task into 4-8 subtask, and specify in terms of objectives. Cover the whole area of interest
- c) Draw out as layered plans, logically & technically correct. None should be missing.

# Stages of a HTA

## 2. Progressing the analysis

- a) Decide on level of detail and stop decomposition.  
Should be consistent between tasks. Can range from detailed to high level description.
- b) Decide if a depth first or breadth first decomposition should be done. Can alternate between the two.
- c) Label and number the HTA.

# Stages of a HTA

## 3. Finalizing the analysis.

- a) Check that decomposition and numbering is consistent. May produce a written account of the processes.
- b) Have a second person look it over. They should know the tasks but not be involved in the analysis.

0 prepare and print  
a letter using  
Microsoft Word

plan 0:: 1-2, then according to requirements 2, 3, 4, 6;  
approximately every 15 minutes,  
following complex procedure,  
prior to printing and prior to quitting-5;  
when session is over-7

